4.3 DATABASE DESIGN

Database Name: chitfund_db
Purpose: Stores relational entities for users, chit plans, KYC, draws, installments, payments, and system logs.
Database Engine: MySQL 8.0 (InnoDB, UTF-8mb4)

Table 1: users

```
json

{

  "id": 1,

  "username": "naresh123",

  "name": "Naresh Kumar",

  "email": "naresh@chitfund.com",

  "phone": "9876543210",

  "password_hash":
"$2a$10$e8wF5Zq7Kj1pNmYxRu3bCeVp4Q.LmDtbC9TzA4nFd8Icmi2",

  "role": "admin",

  "status": "active",

  "aadhar_number": "123456789012",

  "pan_number": "ABCDE1234F",

  "date_of_birth": "1990-05-15",

  "occupation": "Business",

  "kyc_status": "verified",

  "failed_login_attempts": 0,

  "last_login": "2026-01-10T09:00:00",

  "is_deleted": false,

  "created_at": "2026-01-01T10:00:00",

  "updated_at": "2026-01-05T12:00:00"

}
```

Table 2: kyc_documents

```
json

{

  "id": 101,
```

```
  "user_id": 1,
  "document_type": "aadhar_front",
  "file_path": "/uploads/kyc/aadhar_front_1.jpg",
  "status": "verified",
  "rejection_reason": null,
  "verified_by": 2,
  "verified_at": "2026-01-05T10:30:00",
  "created_at": "2026-01-02T15:00:00"
}
```


Table 3: chit_plans


```
json
{
  "id": 201,
  "plan_name": "Gold Plan 2026",
  "plan_type": "gold",
  "chit_type": "lucky_draw",
  "monthly_amount": 5000.0,
  "total_members": 20,
  "current_members": 15,
  "duration_months": 20,
  "total_value": 100000.0,
  "commission_rate": 5.0,
  "net_prize_amount": 95000.0,
  "start_date": "2026-02-01",
  "end_date": "2027-10-01",
  "status": "active",
  "created_by": 1,
  "created_at": "2026-01-10T11:00:00"
}
```


Table 4: plan_members

```
json
{
  "id": 301,
  "user_id": 5,
  "plan_id": 201,
  "ticket_number": "TCK-1024-1",
  "joined_date": "2026-01-15",
  "has_won": false,
  "is_active": true,
  "reason": "personal",
  "priority_score": 5,
  "created_at": "2026-01-15T09:20:00"
}
```

Table 5: monthly_draws

```
json
{
  "id": 401,
  "plan_id": 201,
  "month_number": 1,
  "draw_date": "2026-02-15",
  "chit_value": 100000.0,
  "winning_bid_amount": 25000.0,
  "discount": 25000.0,
  "company_commission": 1250.0,
  "net_discount": 23750.0,
  "dividend_per_member": 1187.5,
  "net_payable": 3812.5,
  "prize_amount": 75000.0,
  "winning_ticket_id": 301,
  "winner_id": 5,
  "status": "completed",
```

```
  "conducted_by": 1,
  "created_at": "2026-02-15T14:00:00"
}
```

Table 6: installments

```
json
{
  "id": 501,
  "plan_member_id": 301,
  "plan_id": 201,
  "month_number": 1,
  "due_date": "2026-02-20",
  "base_amount": 3812.5,
  "penalty_amount": 0.0,
  "total_amount": 3812.5,
  "status": "paid",
  "paid_date": "2026-02-18",
  "payment_method": "UPI",
  "transaction_id": "UPI1234567890",
  "receipt_number": "REC-2026-0001",
  "receipt_path": "/receipts/REC-2026-0001.pdf",
  "verified_by": 1,
  "verified_at": "2026-02-19T10:00:00",
  "created_at": "2026-02-15T15:00:00"
}
```

Table 7: payments

```
json
{
  "id": 601,
  "installment_id": 501,
  "user_id": 5,
```

```
  "amount": 3812.5,
  "payment_method": "UPI",
  "transaction_id": "UPI1234567890",
  "status": "success",
  "notes": "First month installment paid",
  "created_at": "2026-02-18T09:30:00"
}
```

Table 8: promissory_notes

```
json
{
  "id": 701,
  "pronote_id": "PRN-2026-0001",
  "member_id": 5,
  "chit_id": 201,
  "loan_amount": 75000.0,
  "issue_date": "2026-02-16",
  "due_date": "2027-10-01",
  "interest_rate": 0.0,
  "repayment_amount": 75000.0,
  "purpose": "Business expansion",
  "status": "active",
  "signature": "Base64 encoded Canvas digital signature string",
  "created_at": "2026-02-16T10:00:00"
}
```

Table 9: guarantors

```
json
{
  "id": 801,
  "member_id": 5,
  "name": "Ramesh Kumar",
```

```
  "phone": "9988776655",
  "address": "78, South Street, Chennai",
  "relationship_to_member": "Friend",
  "aadhar_number": "987654321098",
  "document_path": "/uploads/guarantors/g_aadhar_1.jpg",
  "created_at": "2026-02-16T10:15:00"
}
```

Table 10: notifications

```
json
{
  "id": 901,
  "user_id": 5,
  "title": "Auction Winner",
  "message": "Congratulations! You have won the auction for Gold Plan 2026.",
  "notification_type": "success",
  "is_read": false,
  "created_at": "2026-02-15T14:30:00"
}
```

Table 11: certificates

```
json
{
  "id": 1001,
  "user_id": 5,
  "plan_id": 201,
  "certificate_number": "CERT-2026-999",
  "file_path": "/certificates/CERT-2026-999.pdf",
  "issued_by": 1,
  "issued_at": "2027-10-05T11:00:00"
}
```

Table 12: reports

```
json

{

  "id": 1101,

  "report_type": "revenue",

  "report_format": "pdf",

  "file_path": "/reports/revenue_feb_2026.pdf",

  "generated_by": 1,

  "generated_at": "2026-02-28T18:00:00"

}
```

Table 13: audit_logs

```
json

{

  "id": 1201,

  "user_id": 1,

  "action": "PAYMENT_VERIFIED",

  "entity_type": "installment",

  "entity_id": 501,

  "details": "Verified payment for REC-2026-0001",

  "ip_address": "192.168.1.100",

  "created_at": "2026-02-19T10:00:05"

}
```

Table 14: system_settings

```
json

{

  "id": 1301,

  "key": "commission_rate",

  "value": "5.0",

  "description": "Default foreman commission percentage",

  "updated_at": "2026-01-01T09:00:00",

  "created_at": "2026-01-01T09:00:00"
```

```
}
```

Table 15: password_resets

```
json

{
  "id": 1401,
  "user_id": 5,
  "reset_token": "a1b2c3d4e5f6g7h8i9j0",
  "expires_at": "2026-03-01T10:00:00",
  "is_used": false,
  "created_at": "2026-03-01T09:00:00"
}
```

Admin Module

- **Secure Administrative Login:** Separate role-based authentication for administrators. JWT Bearer tokens with RBAC middleware restrict all admin-only endpoints. Failed login attempts are tracked and accounts are locked after repeated failures.
- **Member KYC Verification:** Review the uploaded KYC documents (Aadhaar, PAN, address proof, passport photo) for each member. Approve the member to activate their account or reject with a recorded rejection reason. KYC status updates trigger instant member notifications.
- **Chit Plan Management:** Create, configure and activate chit plans by setting plan name, tier (Bronze / Silver / Gold), monthly installment amount, total members, duration, commission rate and start date. The system auto-calculates total pool value and net prize amount.
- **Monthly Auction Conduct:** Open the monthly bidding session, monitor live bids, and trigger the AI Emergency Priority Engine (Rank 1–5) to determine the draw winner. Finalise the auction winner, record the winning bid, and auto-trigger dividend distribution across all active members.

- **Dividend & Commission Calculation:** System enforces the statutory 5% foreman commission cap on each auction discount. Net discount is distributed equally among all plan members, automatically reducing their next monthly installment (net_payable = monthly_amount - dividend_per_member).
- **Installment Verification:** Review submitted UTR references against payment records. Verify or reject installment payments. Verified installments trigger PDF receipt generation (REC-2026-XXXX) and update the member's payment ledger.
- **Guarantor & Pronote Oversight:** View guarantor details submitted by members. Monitor all active Promissory Notes (PRN-2026-XXXX), track their status (pending_signature / active / paid / overdue) and send overdue alerts.

- **Report Generation:** Generate downloadable reports in PDF, CSV or Excel format covering member roster, KYC status, installment ledger, overdue accounts, auction history, revenue summary and full audit log.
- **System Settings:** Configure platform-wide parameters such as commission rate, penalty percentages and notification templates through the system_settings key-value store.

Member Module

- **Account Registration & KYC:** Self-service registration with username, email, phone, Aadhaar and PAN number. Member uploads KYC documents (Aadhaar front/back, PAN card, passport photo, address proof) for admin verification before account activation.
- **Chit Plan Discovery:** Browse available Bronze, Silver and Gold tier chit plans filtered by monthly installment amount, total pool value, duration and current vacancy. Members can view full plan details and net prize amount before enrolling.
- **Plan Enrollment:** One-click enrollment into an active chit plan. System auto-assigns a unique ticket number (TCK-YYYY-N) and records the joining date. A member cannot enroll in the same plan twice (unique constraint on user_id and plan_id).
- **Monthly Installment Payment:** View monthly due statements showing base amount, penalty (if overdue) and total payable after dividend deduction. Settle dues via UPI / QR code scan (GPay, PhonePe) by submitting a 12-digit UTR reference number, or via bank transfer / cash.
- **Auction Bidding:** Participate in the monthly reverse-auction draw. Submit a bid discount amount along with a 1-to-5 emergency priority rank (1 = Medical Emergency, 2 = Education, 3 = Marriage, 4 = Business, 5 = Personal Savings) to compete for the monthly prize pool.

- **Promissory Note Execution:** Upon winning the auction draw, the member receives an auto-generated Promissory Note (PRN-2026-XXXX). The member digitally signs the pronote using the HTML5 Canvas signature pad. The signed agreement is compiled as a downloadable PDF and stored securely.
- **Receipt & Certificate Download:** Download tamper-proof PDF payment receipts (REC-2026-XXXX) for every verified installment and a scheme completion certificate upon chit maturity.
- **Notification Centre:** Receive real-time in-app notifications for auction results, installment due reminders, payment verification updates, and KYC status changes (success / warning / alert / info).

System / Backend Module

- **Authentication & Security Engine:** FastAPI JWT Bearer token issuance and verification using python-jose. BCrypt password hashing via Passlib with unique cryptographic salts. SlowAPI rate-limiting middleware prevents brute-force attacks. All sensitive endpoints require valid role-scoped JWT.
- **AI Emergency Priority Engine:** Automated bidding tie-breaker that ranks competing bids by declared emergency category: Rank 1 (Hospital / Medical) > Rank 2 (Higher Education) > Rank 3 (Marriage / Family) > Rank 4 (Business Capital) > Rank 5 (Personal Savings / Investment). Stored in plan_members.priority_score.
- **UPI / QR Payment Engine:** Dynamically generates UPI QR codes using the Python qrcode library for GPay and PhonePe integration. Validates submitted transaction references against the pattern ^[0-9]{12}$ — invalid UTR format returns HTTP 400 Bad Request.
- **PDF Generation Engine:** ReportLab 5.0.0 compiles legally structured Promissory Note PDFs (PRN-2026-XXXX) embedding the member's Base64-encoded Canvas digital signature. Pillow 12.2.0 generates tamper-proof PDF installment receipts (REC-2026-XXXX) with QR verification codes.
- **Email Notification Relay:** Python smtplib via Google SMTP TLS (Port 587) dispatches automated transactional emails for auction winner announcements, installment due reminders, KYC approval / rejection alerts and account activation confirmations.

- **Audit Log Engine:** Immutable audit_logs table records every critical platform action — LOGIN, KYC_VERIFIED, PAYMENT_VERIFIED, AUCTION_CONDUCTED, PRONOTE_SIGNED — with user ID, entity reference, IP address and timestamp for full regulatory compliance.

Workflow Rules

- **UTR Validation:** Every UPI transaction reference must match the pattern ^[0-9]{12}$ (exactly 12 numeric digits). Non-conforming references are rejected with HTTP 400 Bad Request before any database write.
- **Installment Total Calculation:** Total Payable = Base Installment Amount + Penalty Amount. Net Payable (post-dividend) = Total Payable - Dividend Per Member.
- **Auction Settlement Formula:** Discount = Chit Value - Winning Bid Amount. Company Commission = Discount x 0.05 (5%). Net Discount = Discount - Commission. Dividend Per Member = Net Discount / Total Members.
- **Promissory Note Workflow:** Auto-generated on auction winner selection (status: pending_signature). Activated only after member submits a valid Base64 Canvas digital signature (status: active). Marked paid upon full repayment settlement.